「Def.

A subset $S \subseteq \mathbb{R}^3$ is called a Regular Surface if:

for each $p \in S$, there are open neighbor $V \subset \mathbb{R}^3$ containing p and open $\mathcal{U} \subset \mathbb{R}^2$ with

$X : \mathcal{U} \to V \cap S$ which satisfies

1). X is C^∞.

2). X is Homeomorphism

3). dX_q is 1-1 for each $q \in \mathcal{U}$.

「Thm. If $f : \mathcal{U} \to \mathbb{R}$ is differentiable on an open $\mathcal{U} \subseteq \mathbb{R}^2$,
then $S = \{(x, y, f(x,y)) \mid (x,y) \in \mathcal{U}\}$ is regular surface.

Proof. Enough to show that $X : \mathcal{U} \to S : (u,v) \mapsto (u, v, f(u,v))$ is Parametrization.

For each $q \in \mathcal{U}$, $dX_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ f_x & f_y \end{pmatrix}$ has rank 2 clearly, So 1-1.

X is C^∞ since f is differentiable. Moreover, X is clearly 1-1, so
X^{-1} is also differentiable, Continuous, So proved ☒

Def. Let $U \subseteq \mathbb{R}^n$ be an open set, and let $F: U \to \mathbb{R}^m$ be a mapping.
A point $p \in U$ is called the critical point if: $dF_p: \mathbb{R}^n \to \mathbb{R}^m$ is not onto, and the image $F(p)$ is called critical value.
Otherwise, $q \in \mathbb{R}^m$ is called regular value.

Observation.
For a mapping $f: \mathbb{R}^3 \to \mathbb{R}$, the Jacobian at $p \in \mathbb{R}^3$ is

$$df_p = \begin{bmatrix} f_x(p) & f_y(p) & f_z(p) \end{bmatrix}$$

therefore, df_p is <u>Not</u> onto if and only if $f_x = f_y = f_z = 0$ at p.

This observation gives us that: for $f: U (\subseteq \mathbb{R}^3) \to \mathbb{R}$,
$a \in f[U]$ is critical value if and only if f_x, f_y, f_z are not all zero on the preimage $f^{-1}[\{a\}]$.

[Thm.
If $a \in f[U]$ is regular value of a mapping $f : U (\subseteq \mathbb{R}^3) \to \mathbb{R}$,
then the preimage $f^{-1}[\{a\}] \subseteq \mathbb{R}^3$ is regular surface.
proof. Let $p = (x_0, y_0, z_0) \in f^{-1}[\{a\}]$ be given. Since $a \in f[U]$ is regular value,
f_x, f_y, f_z are not all zero. Without Loss of Generality, assume $f_z(p) \neq 0$,
Define $F : U \to \mathbb{R}^3 : (x,y,z) \mapsto (x, y, f(x,y,z))$. Since f is differentiable, So is F.
And, the Jacobian of F at p is

$$dF_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ f_x & f_y & f_z \end{bmatrix}$$

is invertible, because $f_z(p) \neq 0$, therefore by the inverse function theorem,
for some neighbor V containing p and W containing $F(p)$,
$F : V \to W$ is invertible and $F^{-1} : W \to V : (u,v,t) \mapsto (u, v, g(u,v,t))$ is differentiable.

「Thm. Let $S \subseteq \mathbb{R}^3$ be a regular surface.
Given $p \in S$, there exists a neighbor V containing p such that $V \cap S$ is a graph for some differentiable function either: $\begin{cases} z = f(x,y) \\ y = g(x,z) \\ x = h(y,z). \end{cases}$

Proof. Let $X : \mathcal{U}(\subseteq \mathbb{R}^2) \to S$ be a parametrization of S at p.
Denote $X(u,v) = (x(u,v), y(u,v), z(u,v))$. By the definition, at $q = X^{-1}(p) \in \mathcal{U}$,

$$DX_q = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{bmatrix} \text{ has rank 2,}$$

So, at least one of these is not zero at q:

$$\frac{\partial(x,y)}{\partial(u,v)}, \frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(x,z)}{\partial(u,v)}.$$

Assume $\frac{\partial(x,y)}{\partial(u,v)} \neq 0$. Consider a projection $\pi(x,y,z) = (x,y)$.
Then, $(\pi \circ X)(u,v) = (x(u,v), y(u,v))$.
By the assumption, $D(\pi \circ X)(q) \neq 0$, thus the inverse function theorem gives:
for some neighbor V_1 and V_2 containing q and $(\pi \circ X)(q)$, respectively, $(\pi \circ X) : V_1 \to V_2$ is diffeomorphism. Now, the inverse

$$(\pi \circ X)^{-1} : V_2 \to V_1 : (x,y) \mapsto (u(x,y), v(x,y))$$

is differentiable, now,

$$f : (x,y) \mapsto (x, y, z(u,v))$$

is we desired. ☒

「Ellipsoid.

$$E = \{(x,y,z) \in \mathbb{R}^3 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1\}, \quad (a,b,c \neq 0)$$

Define $f : \mathbb{R}^3 \to \mathbb{R} ; (x,y,z) \mapsto \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$

Then, $E = f^{-1}[\{0\}]$. f is differentiable because it is polynomial,

and $0 \in f[E]$ is a regular point, because $f(x,y,z)=0$ iff $(x,y,z) = \left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right)$

but at this point, $df = (f_x \ f_y \ f_z) \neq 0$, so df is surjective.

Now, by the theorem, E is regular surface

Torus.

Consider a circle S^1 centered at $(0, a, 0)$, with a equation

$$(y-a)^2+z^2=r^2.$$

Rotate this with respect to z-axis, then we obtain a torus

$$z^2=r^2-\left(\sqrt{x^2+y^2}-a\right)^2$$

That is, the torus is; for a map

$$f(x,y,z)=z^2+\left(\sqrt{x^2+y^2}-a\right)^2,$$

$T=f^{-1}[\{r^2\}]$.

f is differentiable excepe $(x,y)\neq(0,0)$, with

$$\frac{\partial f}{\partial x}=2\cdot\left(\sqrt{x^2+y^2}-a\right)^2\cdot x\Big/\sqrt{x^2+y^2}, \quad \frac{\partial f}{\partial y}=2\cdot\left(\sqrt{x^2+y^2}-a\right)^2\cdot y\Big/\sqrt{x^2+y^2},$$

$$\frac{\partial f}{\partial z}=2z$$

「Sphere

Consider a sphere S_+^2 Centered at $(0,0,1)$, and $N=(0,0,2)$. $(x^2+y^2+(z-1)^2=1)$

Define a Stereographic projection

$$\pi : S_+^2\setminus\{N\}\to\mathbb{R}^2 : (x,y,z)\mapsto\left(\frac{2x}{2-z},\frac{2y}{2-z}\right)$$

1). π has an inverse

$$\pi^{-1}:\mathbb{R}^2\to S_+^2\setminus\{N\}:(u,v)\mapsto\left(\frac{4u}{u^2+v^2+4},\frac{4v}{u^2+v^2+4},\frac{2(u^2+v^2)}{u^2+v^2+4}\right)$$

It is deduced by: Consider a straight line

$$l(t)=(0,0,2)+t\cdot[(u,v,0)-(0,0,2)]=(tu,tv,2-2t)$$

Find $l[\mathbb{R}]\cap S_+^2$, as

$$(tu)^2+(tv)^2+(1-2t)^2=1.$$

That is, $t^2u^2+t^2v^2+4t^2-4t+1=1 \Rightarrow t\cdot[t(u^2+v^2+4)-4]=0$

So, $t=0$ or $t=\dfrac{4}{u^2+v^2+4}$

Now, π, π^{-1} both are differentiable and 1-1, it is a Parametrization.

Precisely, $\pi\circ\pi^{-1}=id$, so

$$D\pi\circ D\pi^{-1}=I_2 \Rightarrow D\pi^{-1} \text{ injective} \Rightarrow \text{rank } 2.$$

And, Consider a Monge Patch

$$f:\mathbb{D}\to S_+^2:(x,y)\mapsto\left(x,y,1+\sqrt{1-(x^2+y^2)}\right)$$

Then it covers the North $N\in S_+^2$, Consequently, S_+^2 is regular surface